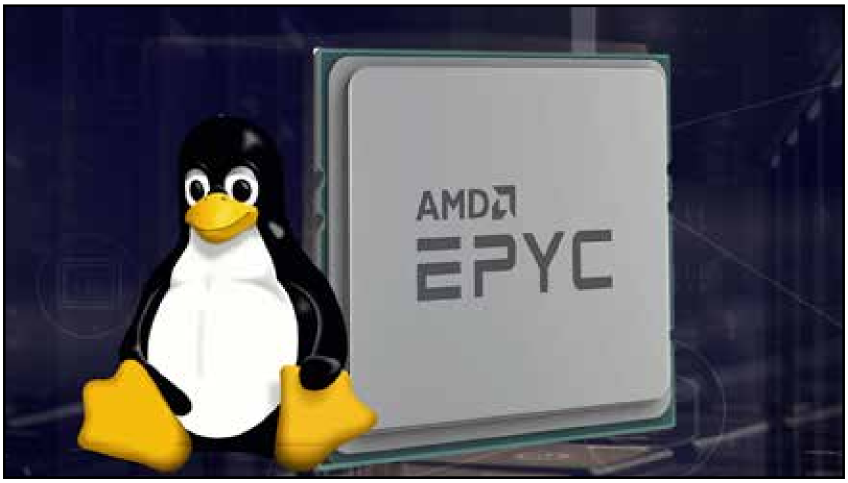
AMD to integrate Smart Data Cache Injection into the Linux kernel



Linux engineers at AMD are working on integrating Smart Data Cache Injection (SDCI) into the kernel, a feature designed for AMD EPYC server processors. SDCI is an innovative capability that enables data to be directly inserted from I/O devices into the CPU's L2/L3 cache.

Recently, a new patch series was released, aiming to prepare the Linux kernel's resource control ("resctrl") functionality for L3 Smart Data Cache Injection Allocation Enforcement (SDCIAE).

This patch series provides details about SDCI and SDCIAE: "Upcoming AMD hardware implements Smart Data Cache Injection (SDCI). Smart Data Cache Injection (SDCI) is a mechanism that enables direct insertion of data from I/O devices into the L3 cache. By directly caching data from I/O devices rather than first storing the I/O data in DRAM, SDCI reduces demands on DRAM bandwidth and reduces latency to the processor consuming the I/O data. The SDCIAE (SDCI Allocation Enforcement) PQE feature allows system software to limit the portion of the L3 cache used for SDCI."

Support for AMD's Smart Data Cache Injection (SDCI) relies on PCI Express TLP Processing Hints (TPH), which were previously discussed on Phoronix when AMD engineers released related patches. These TPHs are hints that help optimise latency and reduce traffic congestion by indicating the best cache location for a Transaction Layer Packet (TLP) when multiple cache locations are available.

The patches suggest that "upcoming" and "new" AMD hardware will support Smart Data Cache Injection. However, SDCI was originally announced as a feature for AMD EPYC Genoa(X) and Bergamo processors. Since that announcement, there has been little news about SDCI until these recent Linux kernel patches. The timing of these patches, just before the launch of AMD EPYC Zen 5 'Turin', raises questions. It is unclear whether this SDCI support is intended for the existing EPYC Bergamo/Genoa(X) processors or only for upcoming platforms, depending on whether there are significant differences in SDCI implementation.

Tails 6.6 improves persistent storage and adds support for newer hardware

The Debian-based Tails (The Amnesic Incognito Live System) has been updated to version 6.6, bringing enhanced hardware support, updated components, and various bug fixes.

Key improvements in Tails 6.6 include better support for newer hardware, such as graphics and Wi-Fi, and the ability to detect additional errors when starting Tails from a USB stick, particularly when resizing the system partition fails. The update also enhances the Additional Software app, making it more stable by preventing crashes when installing virtual packages.

Persistent storage has seen significant improvements in this release. The maximum waiting time when unlocking persistent storage has been increased to 4 minutes, reducing the likelihood of errors. Additionally, issues where the persistent storage settings could freeze after opening documentation links have been resolved.

Tails 6.6 reintroduces the option to enable multiple network interfaces, fixes issues related to connecting to the Tor network using default bridges and improves Tails Cloner by eliminating the 30-second wait time during installation by cloning. This release updates key applications, including Tor Browser to version 13.5.2 and Mozilla Thunderbird to version 115.14.0.

Tails 6.6 includes several smaller bug fixes, which can be reviewed in the full changelog. Users can download Tails 6.6 as an ISO or USB image from the official website or update their existing installations through automatic or manual upgrades.